Week 1: Ground Truth

Option C (Builder) Report

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Repository: https://github.com/aayushd97/ag-news-ground-truth (pull request: https://github.com/aayushd97/ag-news-ground-truth/pull/1)

I labelled 30 AG News items (Zhang, Zhao and LeCun, 2015) by hand before running any model. I then compared my labels, two AI prompts (openai/gpt-oss-20b on Groq, temperature 0) and the official labels. Both prompts matched the official labels on 25 of 30 items and I matched on 23. Every error sits on an item the benchmark labels Business or Sci/Tech. All tables are in the appendix.

Process and disclosures. I committed my rules before any code existed, and my 30 labels before the AI ran. I labelled the items before opening the official labels. The rules come from my own instinct and logic. I had not read the brief's category table or Prompt 1 when I committed the rules. I ran every script myself, and every measured result here comes from outputs saved in the repository's results folder. The work was done on a branch and merged by pull request.

The eight questions

1. Agreement and kappa

Against the official labels: me 23 of 30 (76.7%, kappa 0.69), prompt 1 25 of 30 (83.3%, kappa 0.78), prompt 2 25 of 30 (83.3%, kappa 0.78) (Table A1). Kappa corrects for the agreement expected by chance (Cohen, 1960). All three are above 0.65, though mine sits just under the 0.7 level the brief uses as a warning line. I would show a manager prompt 1's kappa of 0.78 with '25 of 30' beside it, because raw agreement looks better than it should when some categories are easy to get right. I would also tell them that 'right' here means 'matches the benchmark', and the benchmark can be wrong too (question 3). With only 30 items, prompt 1's true agreement rate could plausibly be anywhere from 66% to 93%.

2. My rules against the official rules

The official rules put money first: a company's earnings, a club's finances or a tech firm's share price is Business, and a product's technical capability is Sci/Tech. My rules agreed on money and on Sports, and differed in three places. My World rule claimed any story where politics is the reason behind it, I had no rule for software forecasts, and I did not say what happens when a flaw threatens profit. Those gaps predicted four of my seven misses (Table A3). Items 21 and 23 come from the World clause. Items 22 and 29 come from mixing money and technology. I put a software sales forecast in Business and the benchmark chose Sci/Tech, while I put a phone flaw that hurts profit in Sci/Tech and the benchmark chose Business. Items 18 and 28 were my own inconsistency. Before the reveal, I realised they fit none of my rules, and I kept them anyway. Item 6, a drug safety warning, is explained by neither my rules nor the official ones.

3. Row 21: where I think the benchmark is wrong

Item 21, 'Old Rumors of Gay Sex Prove Powerful on Web', reports that Virginia Republicans chose Del. Thelma Drake to replace Rep. Edward Schrock, who resigned amid allegations. I labelled it World with High confidence, prompt 1 agreed, and the official label is Sci/Tech. **I think the benchmark is wrong for the text supplied.** The story is about a resignation and a successor, which is politics and national affairs. Nothing else fits, as there is no sport, no company or money, and no science or technology. The only link to technology is the last word of the headline, 'Web', where the rumours spread, and the text says nothing about how the web works. The best case for Sci/Tech is that a labeller who saw only the headline filed it by that word. I did not verify how the label was assigned, but one possibility is that it came from the section of the source and not from someone reading the text. I would change my mind if the full article were mainly about how rumours travel online. Prompt 1's agreement is weak evidence, since we read the same snippet, and one row does not show the benchmark is unreliable in general.

4. The confusion matrix

Business and Sci/Tech are the most confused pair for both labellers, with three crossings in my labels and four in prompt 1's (Table A2). The weakest category is Sci/Tech for prompt 1 (recall 4 of 7, F1 0.62) and Business for me (recall 3 of 7, F1 0.55). Accuracy alone would have told me neither. Prompt 1's 83% would be the same whether its five misses were spread across all four categories or packed into two, and it says nothing about direction. My World recall was perfect, but my World precision was only 0.67, because I put four items there that the benchmark files elsewhere.

5. Eighty percent

In the imbalanced set, 24 of 30 items are World news. A system that ignores the text and always answers 'World' gets 24 of 30 right, which is 80%, and would look good on a dashboard. It never finds a single Sports, Business or Sci/Tech story, so for anyone who wants those it is useless. Kappa exposes this, because it subtracts what always guessing the common answer would earn, and the always-World system scores 0.0. My real classifier also scored 80% on this set, but its kappa was 0.52. It found both Sports and both Business items, missed both Sci/Tech items, and only 2 of its 6 Business answers were right. So an 80% score here means very little on its own.

6. My own calibration

My High items were right 20 of 24 times (83%), my Medium items 3 of 5 (60%), and my one Low item (item 18) was wrong. So lower confidence did go with more errors: 3 of my 6 Medium and Low items were wrong, against 4 of 24 High items. But it caught only 3 of my 7 errors, and four were High confidence (items 6, 21, 22 and 29). With one Low item I cannot claim more. The AI's confidence was a better signal: for prompt 1, 22 of 23 answers above 0.95 were right (96%), against 3 of 7 between 0.80 and 0.95 (43%), though seven items is a loose basis.

7. Prompt one against prompt two

The gap is 0 percentage points, because both prompts matched 25 of 30 labels (kappa 0.78 each). They differed on two items. On item 16 (cocoa prices during unrest in Ivory Coast) prompt 1 said World, matching the benchmark, and prompt 2 said Business. On item 28 (a flawed bicycle lock) prompt 1 said Sci/Tech and prompt 2 said Business, matching the benchmark. The errors cancelled each other out. Prompt 1 lists conflict under World and says a product's technical capability is Sci/Tech, which fits its lock answer. Prompt 2 has no boundary rules, so money words ('prices', 'lock manufacturer') plausibly pulled both items to Business. This is only my best guess at the cause. Two items are too few to prove it, because one item is worth 3.3 percentage points. The bigger gap the brief predicted also did not appear. Both prompts ask for the answer in the same JSON format, and neither produced a broken reply, so I could not test how much the format instruction matters.

8. The stack questions

Retrieval. Searching my 30 items for 'world' returned five rows: three official World stories, a Sports story (a World Cup qualifier injury doubt) and a Business story (Siemens, 'the world's fourth-largest maker of mobile phones'). An assistant seeing only these would present a football injury and a profit warning as world news, because keyword matching finds the word and not the meaning.

SQL. The same filter in DuckDB, WHERE lower(text) LIKE '%world%', returned the same five rows as pandas. I would hand the SQL to a colleague because each clause says what it does.

Red run. In the pandas repository, pull request 67463 ended in Failure. The unit-test jobs on Linux for Python 3.11 to 3.14 each finished with 'Process completed with exit code 1', which is what turns a check red. Many other jobs were cancelled because a newer push replaced them. The check warns the author and reviewers, and would stop a merge only if the repository requires it.

Kubernetes. It looks after many copies of an application running on several computers, restarting any that crash and adding or removing copies as demand changes, a bit like a control room.

Other required steps (C2 to C7)

C2. The classifier asks for JSON with a label and a confidence, and counts any invalid reply as a failure without repairing it. The failure rate was 0 of 60 replies on the main set and 0 of 30 on the imbalanced set. That is a result for this model. A different model, or a prompt without the JSON line, could fail more.

C3. A simple word-counting model (TF-IDF with logistic regression) scored 100% on my 30 items when they were inside its training data, and 80% (24 of 30) with them removed. In a second test it scored 96.5% on 6,000 rows it had trained on and 87.8% on 1,600 it had never seen. The honest 80% is one item behind the AI's 25 of 30, so on this sample the AI shows no clear advantage over a cheap model.

C4. The 95% Wilson interval for prompt 1 (25 of 30) is 0.66 to 0.93, so on 3,000 items the agreement could plausibly fall anywhere in that range, a span of 27 points. Thirty items is a weak basis for a claim because each item moves the result by 3.3 points. For a margin of about five points I would want roughly 400 items.

C5. An empty string and 3,000 words of noise both failed quietly, returning status 'ok' with the label World at confidence 0.0 and 0.1 (Table A4). Validation checks a reply's shape, not whether it means anything. I fixed both with guards that return 'empty_input' and 'input_too_long' without calling the API. A Hindi cricket headline was classified correctly as Sports. A story fitting two categories got one confident Business label (0.92), which I did not fix because it comes from the one-label design.

C6. I wrapped the classifier in a Gradio page, launched it with share=True and confirmed the public link works from my phone. The 'Use via API' page shows the endpoint (screenshot in the repository's screenshots folder). The frontend is the web page in the browser. The backend is the Python function on my computer that validates input and calls Groq. The boundary is the HTTP request between them, and the API key stays on the backend side.

C7. I used 30 Airbnb listing descriptions from the group Text Analytics data (four English-speaking cities) and guessed each listing's room type before seeing the host's label. The sample was balanced on purpose (10 entire homes, 10 private rooms, 5 hotel rooms, 5 shared rooms), because about 73% of eligible listings are entire homes. Agreement was 17 of 30 (56.7%, interval 0.39 to 0.73) and kappa was 0.40, well below the 0.7 line (Table A5). I was overconfident: my High items were right 12 of 19 times. The misses came from descriptions of a whole property when the guest rents one room, condo resorts written like hotels, and one misreading of my own (item 1 says 'Shared-Room' and I labelled it Entire). Kappa below 0.7 normally casts doubt on the existing labels, but the host knows what they rent, so the label is probably the more reliable side. The result says descriptions do not reliably reveal room type. Some descriptions do state it outright, and a reader can get those right, but most of my misses were descriptions that did not. For the group project, which compared text-based and metadata-based price models, this suggests text cannot be assumed to carry what room type carries. It rests on 30 listings, and I have not tested it against the project's models.

Limits. Thirty items cannot separate two systems, only I labelled them, the results come from one model at temperature 0, both prompts share the same JSON line, and the imbalanced and C7 samples were chosen deliberately, so their accuracy does not describe real-world data.

References

Cohen, J. (1960) 'A coefficient of agreement for nominal scales', *Educational and Psychological Measurement*, 20(1), pp. 37-46.

Zhang, X., Zhao, J. and LeCun, Y. (2015) 'Character-level convolutional networks for text classification', *Advances in Neural Information Processing Systems*, 28.

Appendix: Tables

Table A1: Agreement and kappa, 30 main items

Comparison	Matches	Agreement	95% Wilson interval	Kappa
Me vs official	23 / 30	0.767	0.59 to 0.88	0.687
AI prompt 1 vs official	25 / 30	0.833	0.66 to 0.93	0.777
AI prompt 2 vs official	25 / 30	0.833	0.66 to 0.93	0.777
Me vs AI prompt 1	26 / 30	0.867	0.70 to 0.95	0.819
Me vs AI prompt 2	25 / 30	0.833	0.66 to 0.93	0.777

Table A2: Confusion matrices (rows = official, columns = predicted)

Me (left) and **AI prompt 1** (right)

Official	World	Sports	Business	Sci/Tech		World	Sports	Business	Sci/Tech
World	8	0	0	0		8	0	0	0
Sports	0	8	0	0		0	8	0	0
Business	2	0	3	2		0	0	5	2
Sci/Tech	2	0	1	4		1	0	2	4

Recall for me: World 1.00, Sports 1.00, Business 0.43, Sci/Tech 0.57. Recall for prompt 1: 1.00, 1.00, 0.71, 0.57.

Table A3: My seven disagreements with the official labels

Item	Story (short)	Mine	Official	Prompt 1	Likely cause
6	Painkiller safety warning	Sci/Tech	Business	Sci/Tech	Not explained by my rules or the table
18	Satellite radio shock jocks	World	Sci/Tech	Business	My inconsistency
21	Virginia Republicans pick a successor	World	Sci/Tech	World	World clause; benchmark likely wrong
22	Software sales forecast	Business	Sci/Tech	Business	Money versus technology
23	Qantas tax treatment	World	Business	Business	World clause
28	Flawed bicycle lock	World	Business	Sci/Tech	My inconsistency
29	Phone flaw may hurt profit	Sci/Tech	Business	Business	Money versus technology

Table A4: C5, trying to break the classifier

Input	Before the fix	After the fix	Verdict before
Empty string	ok, World, 0.0	empty_input, no label	Failed quietly
3,000 words of noise	ok, World, 0.1	input_too_long, no label	Failed quietly
Hindi cricket headline	ok, Sports, 0.99	Unchanged	Handled correctly
Tech firm buys football club	ok, Business, 0.92	Unchanged	Invented certainty

Table A5: C7, host room type against my guess (rows = host label)

Host label	Entire	Hotel	Private	Shared
Entire home/apt	6	2	2	0
Hotel room	2	3	0	0
Private room	5	1	4	0
Shared room	1	0	0	4